Griffin T. Goodwin

Machine Learning & Data Science — Atlanta, Georgia

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EDUCATION

5/2022 – Present **Georgia State University** — Atlanta, GA
Ph.D. Candidate in Astronomy – **NASA FINESST Fellow, 2026 Hadley Award Recipient**
Master of Science in Physics (2024) GPA: 4.19/4.30

8/2018 – 5/2022 **Georgia Institute of Technology** — Atlanta, GA
Bachelor of Science in Physics
Concentration in Astrophysics, Minor in Computational Data Analysis GPA: 3.94/4.00

EXPERIENCE

6/2025 – Present **HelioLab Researcher** — Frontier Development Lab – Trillium, NASA, NVIDIA, Google
Advisors: A. Vourlidas, R. Jarolim, V. Sadykov, C. Schirninger, L. Pratt
Collaborated with an interdisciplinary team of data scientists and heliophysicists to address critical challenges in space weather through machine learning (ML). Developed a Vision Transformer-based regression and localization model using a large-scale dataset of over 200,000 extreme ultraviolet (EUV) solar images (3,300+ hours of observations). The model accurately identifies the locations and intensities of individual solar flares, achieving a 79% reduction in root mean squared error over a simple baseline. Our data and model are publicly deployed on Hugging Face Spaces, enabling real-time inference via a web interface. Read more about this project at my Frontier Development Lab blog post here.

5/2022 – Present **Graduate Student – SunSET Lab** — Georgia State University
Advisors: V. Sadykov, D. Kempton, P. Martens
Awarded the competitive NASA FINESST Fellowship to develop ML models that forecast solar flares—a highly imbalanced, rare-event classification problem—from magnetogram and EUV time-series data. Quantified how training-data volume and quality affect predictive performance across MLP, SVM, and ensemble models. Currently leading SWAN-SF 2.0, a terabyte-scale, machine-learning-ready benchmark dataset developed on Apache Spark, featuring an automated pipeline for ingestion, preprocessing, and feature engineering that continuously updates as new data is received.

8/2021 – 5/2022 **Undergraduate Research Assistant** — Georgia Institute of Technology
Advisor: J. Sowell
Performed multi-band time-series analysis of photometric light curves (R, B, V, TESS) to derive physical parameters of an eclipsing binary system.

6/2021 – 8/2021 **National Science Foundation Physics REU** — Rensselaer Polytechnic Institute
Advisor: H. Newberg
Performed statistical analysis of a large astronomical survey (DES DR2) to characterize stellar-brightness distributions across eight Milky Way halo globular clusters.

SELECTED PUBLICATIONS

05/2026 **Improving Solar Flare Soft X-ray Classification With FOXES** — ApJ
Goodwin, G.T. et al., The Astrophysical Journal

10/2025 **FOXES: A Framework For Operational X-ray Emission Synthesis** — arXiv
Goodwin, G.T. et al. – Accepted to the ML4PS Workshop at NeurIPS

3/2025 **The Impacts of Magnetogram Projection Effects on Solar Flare Forecasting** — ApJ
Goodwin, G.T. et al., The Astrophysical Journal

3/2024 **Investigating Performance Trends of Simulated Real-time Solar Flare Predictions** — ApJ
Goodwin, G.T. et al., The Astrophysical Journal

TECHNICAL SKILLS

Languages: **Python**, **SQL**, Julia, C, Java

ML / Deep Learning: **Scikit-learn**, **PyTorch**, TensorFlow, Keras

Methods: **Time-Series Analysis**, **Vision Transformers**, Statistical Modeling, CNNs, Ensemble Methods

Data & Analysis: **Pandas**, **NumPy**, **Matplotlib**, **SunPy**, SciPy, Seaborn

Tools & Infrastructure: **Git**, **Linux**, **Google Cloud Platform**, Spark, Docker, Hugging Face, Jupyter, HPC / Slurm, Tableau, LaTeX

GRANTS, FELLOWSHIPS, & AWARDS

2026 **Hadley Award**, Outstanding Advanced Graduate Student in Physics — Georgia State University

2023 – 2026	**NASA FINESST Fellow**, $150,000 Research Grant	
2025	**ML-Helio Travel Grant**	
2023, 25, 26	**Space Weather Workshop Travel Aid**	
2023, 24, 26	**SHINE Student Travel Aid**	
2022 – 2023	**Second Century Initiative Fellow**	Georgia State University
2018 – 2022	**Dean's List**	Georgia Institute of Technology

SELECTED PROJECTS

12/2025 – Present **SOL.SWx** — App Store

Currently developing an iOS app that integrates live data from NASA and NOAA public APIs to monitor current space weather conditions (150+ downloads).

5/2022 – Present **Data Mining Lab Member** — Georgia State University

Research group of computer scientists and astronomers focused on space weather prediction through ML.

5/2022 – 12/2022 **Outer-Solar System Body Classification** — ASTR 8850 – Georgia State University

Developed decision tree, SVM, and neural network classifiers for categorizing Jupiter trojans, centaurs, and TNOs.

1/2022 – 5/2022 **MARTAVIZ: Transit Visualization** — CX 4242 – Georgia Institute of Technology

Node.js visualization tool for MARTA bus arrival times, rated higher than official app by 15-person user study.

1/2021 – 5/2021 **Stroke Prediction Project** — CX 4240 – Georgia Institute of Technology

Developed Naïve Bayes, logistic regression, SVM, and decision tree models achieving 75% stroke prediction accuracy.

TEACHING & MENTORING

2023 – Present	**Graduate Mentor**, CS M.S. student — weekly 1:1s, code review, and technical direction	Georgia State University
2022 – 2024	**Teaching Assistant**, Introductory Astronomy Labs	Georgia State University

SELECTED TALKS & PRESENTATIONS

9/2025 **Machine Learning in Heliophysics** — Madrid, Spain

Poster: *FOXES: A Framework for Operational X-ray Emission Synthesis*

6/2025 **Solar Heliospheric and Interplanetary Environment** — Charleston, SC

Session Organizer: Intertwining Physics-Based Simulations and ML in Heliophysics

3/2025 **Space Weather Workshop** — Boulder, CO

Selected Lightning Talk: *Assessing the Impacts of Magnetogram Projection Effects on Solar Flare Forecasting*

10/2024 **Annual International AL Plasma Physics Conference** — Huntsville, AL

Invited Talk: *The Data Mining Lab at GSU: Harnessing Big Data and AI for Solar Transient Event Forecasting*